KARMUKILAN

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RESEARCH INTEREST

- Theoretical Condensed Matter and Statistical Physics
- Dynamical Systems Theory
- Mathematical physics
- Functional Analysis
- Mathematical Modelling of Biological Systems
- Calculus of Variations
- Analysis on Topological spaces

EDUCATION

CURRENT: INTEGRATED MSc. **(August 2023 - Present)**
Third-year Integrated MSc.
National Institute of Science Education and Research, Bhubaneswar, Odisha, India
With 8.58 CGPA

HIGHER SECONDARY EDUCATION **(2023)**
CS Academy ,Coimbatore, Tamil Nadu, India
With 90% in the CBSE board Examination

SECONDARY SCHOOL EDUCATION **(2021)**
Maharishi Vidya Mandir Senior Secondary School, Kumbakonam, Tamil Nadu, India
With 94% in the CBSE board Examination

RESEARCH EXPERIENCE

Summer Internship (2024) on "**Non-linear Dynamics and Mathematical modelling**" under the guidance of **Dr.Chandralaka Meena,IISER Thiruvananthapuram,India**

During the project I familiarized myself with the formalism associated with Dynamical systems like Bifurcations of fixed points, perturbation analysis, phase portraits and linear approximations of non-linear systems about fixed points.I was able to reproduce the numerical simulations for soliton solutions of schrodinger equation with cubic potential dependence.

Winter Internship(2024) on "**Kuramoto model**" under the guidance of **Dr.V K Chandrasekhar, SASTRA, Thanjavur, India**

The goal of the project was to understand the structure of the Kuramoto model in the finite case and with the thermodynamic limit. I was able to prove the stability of the incoherent phase for the mean field kuramoto model under the thermodynamic limit. Plots of the complex order parameter vs coupling parameter for the finite oscillators case.

Summer Internship(2025) on “**Basic Functional Analysis**“ under the guidance of **Dr.Arup Chattopadhyay, IIT Guwahati,India**

The goal of the project was to understand the Analytical structures that can be imposed on Infinite dimensional vector spaces especially Banach spaces. Extensively studied the characterization of bounded linear maps on Banach spaces using Hahn-Banach Theorem,Uniform bounded-ness principle and the open mapping theorem. Topological structures on Operator spaces were also explored.

Summer Internship(2025) on “**Integration theory and fixed point theory**“ under the guidance of **Dr.Rajesh , IIT Tirupati, India**

The two main goals of the project was to impose a variational structure on Partition space of compact real interval and to study Brouwer’s fixed point theorem. A continuous function on the interval allows us to define a scalar functional on the space which allows us to imbue a variational structure on the partition spaee. Proved the brouwer’s fixed point theorem using Sperner’s lemma.

SCHOLARSHIPS

Recipient of the “**DISHA DAE SCHOLARSHIP**” program sponsored and managed by the Department of Atomic energy, Government of India.

SKILLS

Programming languages: **Python**
Document typesetting: **LaTeX**
Mathematical modelling and analysis
Software: MS word,MS excel,MS powerpoint
Creative thinking and Problem solving

RELEVANT COURSE WORK

Integrated MSc., Course Work

- Introductory Physics 1,2
- Real Analysis 1,2
- Metric Space Theory
- Calculus of Several Variables
- Complex Analysis
- Measure Theory
- Probability Theory
- Differential Equations
- Graph Theory
- Quantum Mechanics 1
- Statistical Mechanics
- Condensed Matter Physics
- Mathematical Methods in Physics 1
- Group Theory
- Rings and Module theory
- Field Theory
- Number Theory

Github link to access the project works :https://github.com/Karmukilan-S